lifted off the Earth by a nonreusable *Titan/Centaur* booster configuration. Weighing about a ton, a *Voyager* would fill a small house. Each draws about 400 watts of power—considerably less than an average American home—from a generator that converts radioactive plutonium into electricity. (If it had to rely on solar energy, the available power would diminish quickly as the ship ventured farther and farther from the Sun. Were it not for nuclear power, *Voyager* would have returned no data at all from the outer Solar System, except perhaps a little from Jupiter.)

The flow of electricity through the innards of the spacecraft would generate enough magnetism to overwhelm the sensitive instrument that measures interplanetary magnetic fields. So the magnetometer is placed at the end of a long boom, far from the offending electrical currents. With other projections, it gives *Voyager* a slightly porcupine appearance. Cameras, infrared and ultraviolet spectrometers, and an instrument called a photopolarimeter are on a scan platform that swivels on command so these devices can be aimed at a target world. The spacecraft must know where Earth is if the antenna is to be pointed properly and the data received back home. It also needs to know where the Sun is and at least one bright star, so it can orient itself in three dimensions and point properly toward any passing world. If you can't point the cameras, it does no good to be able to return pictures over billions of miles.

Each spacecraft cost about as much as a single modern strategic bomber. But unlike bombers, *Voyager* cannot, once launched, be returned to the hangar for repairs. The ship's computers and electronics are therefore designed redundantly. Much key machinery, including the essential radio receiver, had at least one backup—waiting to be called upon should the hour of need ever arrive. When either *Voyager* finds itself in trouble, the computers use branched contingency tree logic to work out the appropriate course of action. If that doesn't work, the ship radios home for help.

As the spacecraft journeys increasingly far from Earth, the round-trip radio travel time also increases, approaching eleven hours by the time *Voyager* is at the distance of Neptune. Thus, in case of emergency, the spacecraft needs to know how to put itself into a